

Tulsi Rajnikant Patel

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EDUCATION

Arizona State University

Masters in Computer Science – GPA 4.00

August 2023 – May 2025

Tempe, AZ

Nirma University

Bachelors of Technology in Computer Science and Engineering | Minor in Finance

July 2019 – May 2023

Ahmedabad, India

WORK EXPERIENCE

Machine Learning Engineer

Alkermes Inc.

March 2026 – Present

Waltham, MA (Remote)

- Developing scalable EEG-based treatment prediction **machine learning** models (Elastic Net, Random Forest, LightGBM) with feature engineering, PCA-based dimensionality reduction, evaluation pipelines and visualizations.

Machine Learning Engineer Intern

Alkermes Inc.

July 2025 – February 2026

Waltham, MA

- Spearheading development of a clinical report automation **Streamlit** app using **Snowflake AI Cortex**, featuring an **AI chat assistant** that lets medical writers refine section content via **prompts**, reducing manual drafting effort by 80%.
- Architected secure **RAG pipelines** using Snowflake Vector Store and **AWS S3**, enabling efficient retrieval and prompt orchestration for report generation with >90% summary accuracy.
- Integrated **Large Language Models** (GPT-5, Llama-3.1-70B, Mistral Large) into AI pipelines to generate 10+ section summaries and containerized the application using **Docker** for deployment on **Snowpark Container Services**.

Research and Development Intern

Sahajanand Technologies Private Limited

January 2023 – May 2023

Surat, India

- Developed a **Flask** web application for automated diamond inclusion detection, storing grading metadata in **DynamoDB** to enable scalable retrieval and analysis.
- Prototyped defect detection systems using **contour-based** and **segmentation** techniques in **Python**, improving inclusion identification in diamond images to 93% accuracy.
- Customized **image-processing pipelines** with **OpenCV** and **Scikit-image** to extract facet and table zones, improving region segmentation and reducing manual inspection overhead.

Undergraduate Research Intern

Nirma University

June 2022 – December 2022

Ahmedabad, India

- Delivered interactive **PowerBI** dashboards showcasing gender-based language patterns, contributing to a university-funded digital humanities research **project**.🔗
- Orchestrated a large-scale tweet analysis pipeline collecting 600k+ multilingual tweets from 523 politicians using **SNScrape**, profile-based gender tagging, and **Google Translate API**.
- Configured and trained **deep learning** models (CNN, ANN, LSTM, GRU, RNN) and **LLMs** (BERT, RoBERTa) to classify sentiment and gendered language in political discourse, achieving up to 86% accuracy.

Software Engineer Intern

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June 2022 – August 2022

Surat, India (Remote)

- Co-developed backend modules for “CleanZone”, a full-stack laundry logistics platform, enabling real-time product tracking and OTP-based authentication using **Node.js**, **MongoDB**, and **2Factor API**.
- Implemented QR code generation and password recovery in **JavaScript**, streamlining product handoff and reducing manual tracking errors across partner operations.

PROJECTS

Bon Appetit - A Restaurant Website

- Programmed a full-stack website with **React**, **Node.js** and **MongoDB**, supporting 100+ menu items and reservations.
- Designed a **Bootstrap UI** with interactive components, boosting engagement and processing 200+ monthly bookings.
- Secured admin panel with **JWT authentication** and configured email notifications via **Nodemailer API**.

AI - Powered Clinical Assistant

- Programmed a **RAG** assistant with OpenAI’s **text-embedding-3-small**, semantically indexing 60K+ clinical records from MedQA dataset and deploying 1.2M+ embeddings via **MongoDB Atlas** to reduce response time.
- Leveraged **Mistral-7B-Instruct** via **vLLM** to develop HIPAA-compliant responses, accomplishing 91.5% accuracy.

SKILLS

Programming Languages: Python, C/C++, Java, Swift, SQL, HTML, CSS, JavaScript, Golang

Databases and Cloud Infrastructure: AWS, Docker, Kubernetes, MySQL, PostgreSQL, MongoDB, Git, Spark

Frameworks: Hugging Face, LangChain, TensorFlow, PyTorch, OpenCV, Scikit-image, Django, Node.js, React

Courses: Artificial Intelligence, Cloud Computing, Computer Networking, Data Structures and Algorithm, Database Management, Generative AI, Machine Learning, Natural Language Processing, Software Engineering, Web Technologies